Chapter 16, Problem 12PP

Problem

Develop a set of state variable equations that represent the following differential equation.

$$\frac{d^3y}{dt^3} + 18\frac{d^2y}{dt^2} + 20\frac{dy}{dt} + 5y = z(t)$$

Step-by-step solution

Step 1 of 3

Given differential equation is

$$\frac{d^3y}{dt^3} + 18\frac{d^2y}{dt^2} + 20\frac{dy}{dt} + 5y = z(t)$$

We select the state variables,

Let $x_1 = y(t)$, therefore

$$\dot{x}_1 = \dot{y}(t) \qquad ---(1)$$

Now

$$x_2 = \dot{x}_1 = \dot{y}(t)$$

Step 2 of 3

We are looking at the second order system that would normally have two first-order terms in solution.

We have

$$\dot{x}_2 = \ddot{y}(t) \qquad ---(2)$$

And where

$$\Rightarrow x_3 = \dot{x}_2 = \ddot{y}(t)$$

Now at the third-order system can be taken as

We have

$$\dot{x}_3 = \dddot{y}(t)$$

Step 3 of 3

$$\dddot{y}(t)+18\ddot{y}(t)+20\dot{y}(t)+5y(t)-z(t)=0$$

$$\dot{x}_3=-18x_3-20x_2-5x_1+z(t)\quad ---(3)$$

From equation (1), (2) and (3), we can write in matrix equations form are

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -5 & -20 & -18\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}0\\ 0\\ 1\end{bmatrix}z(t)$$

$$y(t)=\begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

Therefore, by the state variable equations the matrices are

$$\boxed{\begin{array}{l}A=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -5 & -20 & -18\end{bmatrix}\\ B=\begin{bmatrix}0\\ 0\\ 1\end{bmatrix}\\ C=\begin{bmatrix}1 & 0 & 0\end{bmatrix}\end{array}}$$